HAna Exam Sheet 2025/26 - Back From The Dead

Remark: Notation

- $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$

Theorem: Random Identities

- $\sum_{n=1}^{\infty} 2^{-n} \epsilon = \epsilon$ for $\epsilon > 0$.

Theorem: Convergence Tests

Reals, Sequences, and Series

Real Sequences

Definition: Bounds and Boundedness

Let $A \subseteq R$.
* $x \in \mathbb{R}$ is an **upper bound for** A if $x \geq a$ for all $a \in A$
* A is **bounded above** if there exists an upper bound $x \in \mathbb{R}$ for A.
* $s \in \mathbb{R}$ is a **least upper bound** or **supremum** for A if s is an upper bound for A, and $x \geq s$ whenever x is an upper bound for A.
* **Lower bounds** stuff is akin to this - if u don't know it go back to FPM
* A is **bounded** if it is bounded above and below

Axiom: The Completeness Axiom

Every bounded above non-empty $A \subseteq \mathbb{R}$ has a least upper bound in $\mathbb{R}$

Definition: Nested Sets

A sequence $(I_n)_{n \in \mathbb{N}}$ of sets is **nested** if $I_n \supseteq I_{n+1}$ for all $n \in \mathbb{N}$

Theorem: The Nested Interval Theorem

If $(I_n)_{n \in \mathbb{N}}$ is a nested sequence of nonempty closed bounded intervals, then the set $E := \bigcap_{n \in \mathbb{N}} I_n = \{x \in \mathbb{R} : x \in I_n \forall n \in \mathbb{N}\}$ is nonempty. If $\lambda(I_n) \to 0$ as $n \to \infty$ then E contains exactly one element.

Definition: Convergence

A real sequence $(a_n)_{n \in \mathbb{N}}$ **converges** to $a \in \mathbb{R}$ iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$. A sequence which does not converge to some real number **diverges**.
- We say $a_n \to \infty$ as $n \to \infty$ if $\forall R \in \mathbb{R}, \exists N \in \mathbb{N}$ s.t. $a_n > R$ for all $n \geq N$
- We say $a_n \to -\infty$ as $n \to \infty$ if $\forall R \in \mathbb{R}, \exists N \in \mathbb{N}$ s.t. $a_n < R$ for all $n \geq N$

Definition: Monotonicity

Let $(a_n)_{n \in \mathbb{N}}$ be a real sequence and $A := \{a_n : n \in \mathbb{N}\}$. Then:
(1) $(a_n)_{n \in \mathbb{N}}$ is **monotone non-decreasing** if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$
(2) $(a_n)_{n \in \mathbb{N}}$ is **monotone non-increasing** if $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$
(3) $(a_n)_{n \in \mathbb{N}}$ is **bounded above/bounded below/bounded** if A is such
MCT For Seqs: If a real sequence $(a_n)_{n \in \mathbb{N}}$ is monotone non-decreasing and bdd above; or monotone non-increasing and bdd below; it converges.

Major Definition: Cauchy Sequences

A real sequence $(a_n)_{n \in \mathbb{N}}$ is **Cauchy** if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|a_n - a_m| < \epsilon$ for all $n, m \geq N$.

Major Theorem: The Cauchy Criterion

A real sequence is convergent if and only if it is Cauchy.
Proof. 1: Let $(a_n)_{n \in \mathbb{N}}$ be convergent. Then $\exists a \in \mathbb{R}$ s.t. $\lim_{n \to \infty} a_n = a$, and for all $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $|a_n - a| < \epsilon/2$ when $n \geq N$. Then by the TI, $|a_n - a_m| = |(a_n - a) + (a - a_m)| \leq |a_n - a| + |a - a_m| = |a_n - a| + |a_m - a| < \epsilon/2 + \epsilon/2 = \epsilon$ for $m, n \geq N$, thus the sequence is Cauchy.
2: Let $(a_n)_{n \in \mathbb{N}}$ be Cauchy. First show it is bdd. Let $\epsilon = 1$. Then $\exists N \in \mathbb{N}$ s.t. $|a_n - a_m| < 1$ for $m, n \geq N$. In particular $|a_m| < 1 + |a_N|$ for all $m \geq N$, so $|a_m| \leq \max\{|a_1|, \ldots, |a_{N-1}|, |a_N| + 1\}$ for $m \in \mathbb{N}$, so $(a_n)_{n \in \mathbb{N}}$ is bdd. Secondly, by the B-WT, $\exists (a_{n_k})_{k \in \mathbb{N}}$ s.t. $a_{n_k} \to a \in \mathbb{R}$ as $k \to \infty$. Finally, show $a_n \to a$: Let $\epsilon > 0$. Since $(a_n)_{n \in \mathbb{N}}$ is Cauchy, $\exists N, K \in \mathbb{N}$ s.t. $|a_n - a_m| < \epsilon/2$ for $m, n \geq N$ and $|a_{n_k} - a| < \epsilon/2$ for $k \geq K$. Fix $k \in \mathbb{N}$ s.t. $k \geq K$ and $n_k \geq N$. Then $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \epsilon/2 + \epsilon/2 = \epsilon$, $\forall n \geq N$. Thus $a_n \to a$ as $n \to \infty$ as required. □

Definition: Subsequences

Let $(a_n)_{n \in \mathbb{N}}$ be a real sequence. A **subsequence** of $(a_n)_{n \in \mathbb{N}}$ is a real sequence of the form $(a_{n_k})_{k \in \mathbb{N}}$ where for each $k \in \mathbb{N}$ there exists $n_k \in \mathbb{N}$ such that $n_1 < n_2 < \ldots < n_k < n_{k+1} < \ldots$. Ie, a subsequence is just some (possibly all) elements of a sequence taken in order.

Theorem: Bolzano-Weierstrass Theorem

Every bounded sequence of real numbers has a convergent subsequence.
Proof. Let $(a_n)_{n \in \mathbb{N}}$ be a bdd seq., with upper/lower bound $a, b \in \mathbb{R}$. Let $I_1 = [a,b]$ and divide I_1 into $I_1' = [a,(a+b)/2], I_1'' = [(a+b)/2, b]$. Since $I_1 = I_1' \cup I_1''$, at least one of I_1', I_1'' contains infinitely many a_n. Choose one with this property and denote it I_2. Let $n_2 \in \mathbb{N}$ be s.t. $n_2 > n_1 = 1$ and $a_{n_2} \in I_2$. Repeat inductively. By the above process we recursively define a nested sequence of intervals $(I_k)_{k \in \mathbb{N}}$ and subsequence $(a_{n_k})_{k \in \mathbb{N}}$ s.t. each I_k contains infinitely many terms of $(a_n)_{n \in \mathbb{N}}$, $a_{n_k} \in I_k$, and $\lambda(I_k) = 2^{-(k-1)}(b-a)$. Suppose in this way we construct $a_{n_1}, \ldots, a_{n_k}, k \in \mathbb{N}$, and intervals $I_1, \ldots, I_k$. Divide I_k into I_k', I_k'' as above, again at least one contains infinitely many a_n. Choose one with this property, denote it I_{k+1}. Let $n_{k+1} \in \mathbb{N}$ be s.t. $n_{k+1} > n_k$, $a_{N_{k+1}} \in I_{k+1}$.
Now observe $(I_k)_{k \in \mathbb{N}}$ is a nested seq. of clounded intervals, so by the NIT there is $x \in \mathbb{R}$ s.t. x lies in all I_k. Since $a_{n_k} \in I_k$, $|x - a_{n_k}| \leq \lambda(I_k) = \frac{b-a}{2^{k-1}}$ for all $k \in \mathbb{N}$. Then by the Squeeze Theorem, $a_{n_k} \to x$ as $k \to \infty$. □

Series

Major Definition: Infinite Series

Let $(a_k)_{k \in N}$ be a real sequence. We form the **sequence of partial sums** $(s_n)_{n \in \mathbb{N}}$ by $s_n := \sum_{k=1}^{n} a_k$ for all $n \in \mathbb{N}$.
- $\sum_{k=1}^{\infty} a_k$ **converges to** a if $\lim_{n \to \infty} s_n$ exists and $= a$, else **diverges**.

Major Theorem: The Cauchy Criterion For Series

Let $(a_k)_{k=1}^{\infty}$ be a real sequence. Then $\sum_{k=1}^{\infty} a_k$ converges iff for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\left|\sum_{k=m+1}^{n} a_k\right| < \epsilon$ for all $n > m \geq N$.
Proof. Let $(s_n)_{n \in \mathbb{N}}$ be the seq. of partial sums of the series. Observe that $s_n - s_m = \sum_{k=1}^{n} a_k - \sum_{k=1}^{m} a_k = \sum_{k=m+1}^{n} a_k$ for $n > m$. Then the series satisfies the Cauchy criterion iff $(s_n)_{n \in \mathbb{N}}$ is Cauchy. $(s_n)_{n \in \mathbb{N}}$ is Cauchy iff it is convergent, and convergent iff $\sum_{k=1}^{\infty} a_k$ converges. □

Definition: Absolute Convergence

Let $(a_n)_{n \in \mathbb{N}}$ be a real sequence.
(1) We say $\sum_{n=1}^{\infty} a_n$ **converges absolutely** if $\sum_{n=1}^{\infty} |a_n|$ converges
(2) We say $\sum_{n=1}^{\infty} a_n$ **converges conditionally** if it converges but $\sum_{n=1}^{\infty} |a_n|$ diverges.
Absolute convergence implies convergence.
Proof. Let $(a_k)_{k=1}^{\infty}$ be a real seq., and $\sum_{n=1}^{\infty} a_n$ be absolutely convergent. We show the series satisfies the Cauchy criterion: Let $\epsilon > 0$. Since $\sum_{n=1}^{\infty}$ cnvges absolutely, by the Cauchy criterion for series $\exists N \in \mathbb{N}$ s.t. $\sum_{k=m+1}^{n} |a_k| < \epsilon$ for all $n > m \geq N$. Now let $n > m \geq N$ and apply the TI repeatedly to obtain $\left|\sum_{k=m+1}^{n} a_k\right| \leq \sum_{k=m+1}^{n} |a_k| < \epsilon$. This establishes the Cauchy criterion for $\sum_{n=1}^{\infty} a_n$, so the series is convergent. □

Theorem: The Riemann Rearrangement Theorem

Let $(a_n)_{n \in \mathbb{N}}$ be a real sequence and suppose $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. Then for *any* $s \in \mathbb{R}$, there exists a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that the rearrangement $\sum_{n=1}^{\infty} a_{\sigma(n)}$ converges to s.

Note the RRT does not apply to absolutely convergent series. Indeed, for $\sum_{n=1}^{\infty} a_n$ absolutely convergent, and any bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is convergent and $\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$

Subsets Of The Real Line

Definition: Open Sets

We say a set $U \subseteq \mathbb{R}$ is **open** if for all $x \in U$ there exists $r > 0$ such that $(x-r, x+r) \subseteq U$.
- The union of any collection of open sets is open
- The intersection of any finite collection of open sets is open
- Any nonempty open $U \subseteq \mathbb{R}$ is a finite or countable union of nonempty disjoint open intervals.
- If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) > \lambda\}$ is open.

Definition: Closed Sets

A set $F \subseteq \mathbb{R}$ is **closed** if its complement $\mathbb{R} \backslash F$ is open.
- The union of any finite collection of closed sets is closed
- The intersection of any collection of closed sets is closed
- If $f : \mathbb{R} to \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) \leq \lambda\}$ is closed.
- If $F \subseteq \mathbb{R}$ closed and $U \subseteq \mathbb{R}$ open; $U \backslash F$ is open and $F \backslash U$ is closed

Definition: Sequential Closure

$F \subseteq \mathbb{R}$ is **sequentially closed** if for every convergent sequence $(x_n)_{n \in \mathbb{N}}$ s.t. $x_n \in F$ for all $n \in \mathbb{N}$, the limit $x := \lim_{n \to \infty} x_n$ satisfies $x \in F$. $F \subseteq \mathbb{R}$ is closed if and only if it is sequentially closed.
Proof. First prove by contrapositive every closed set is sequentially closed: Let F be closed and $(x_n)_{n \in \mathbb{N}}$ convergent s.t. $x := \lim_{n \to \infty} x_n \notin F$. We show $\exists N \in \mathbb{N}$ s.t. $x_N \notin F$. Since F is closed, by definition $U := \mathbb{R} \backslash F$ is open. Moreover $x \in U$, so $\exists \epsilon > 0$ s.t. $(x - \epsilon, x + \epsilon \subseteq U)$, and $N \in \mathbb{N}$ s.t. $|x_n - x| < \epsilon$ for all $n \in \mathbb{N}$. In particular, $x_N \in (x - \epsilon, x + \epsilon) \subseteq U$, so $x_N \notin F$.
Now prove by contradiction every sequentially closed set is closed: Suppose F is not closed, equivalently $U := \mathbb{R} \backslash F$ is not open. Then $\exists x \in U$ s.t. $\forall r > 0$, $(x - r, x + r) \cap F \neq \emptyset$. For $n \in \mathbb{N}$, let x_n be any element of $(x - \frac{1}{n}, x + \frac{1}{n}) \cap F$. Then $(x_n)_{n \in \mathbb{N}}$ converges to x, since $|x - x_n| \leq \frac{1}{n}$. In particular, since F is sequentially closed, we must have $x \in F$, a contradiction. □

Definition: Level Sets

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and $\lambda \in \mathbb{R}$:
- The **level set** $\{x \in \mathbb{R} : f(x) = \lambda\}$ is closed
- The **strict superlevel set** $\{x \in \mathbb{R} : f(x) > \lambda\}$ is open
- The **superlevel set** $\{x \in \mathbb{R} : f(x) \geq \lambda\}$ is closed
- The **strict sublevel set** $\{x \in \mathbb{R} : f(x) < \lambda\}$ is
- The **sublevel set** $\{x \in \mathbb{R} : f(x) \leq \lambda\}$ is closed

Then for continuous $f : \mathbb{R} \to \mathbb{R}$ and $a, b \in \mathbb{R}, a < b$, we can characterise the level set $\{x \in \mathbb{R} : a \leq f(x) < b\}$ as the intersection between a closed set and an open set.

Definition: The Middle Thirds Cantor Set

The **Middle Thirds Cantor Set** C is a closed set defined by the following recursive process:
(1) Divide [0,1] into thirds, remove the open middle third interval $(\frac{1}{3}, \frac{2}{3})$.
(2) Divide the remaining two intervals into thirds and remove the open middle thirds intervals $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$.
(3) Repeat this process ad infinitum.
At the nth stage of the construction we have 2^n closed intervals $(I_{n,k})_{k=1}^{2n}$ with $\lambda(I_{n,k}) = 3^{-n}$ for all $1 \leq k \leq 2^n$. Then $C := \bigcap_{n=1}^{\infty} C_n$ where $C_n := +\bigcup_{k=1}^{2^n} I_{n,k}$ for $n \in \mathbb{N}$. Each $I_{n,k}$ is a closed interval so their finite union C_n is closed, and C, the intersection of these closed C_n, is also closed.

Uniform Continuity, Uniform Convergence, Power Series

Uniform Continuity

Major Definition: Uniform Continuity

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. f is **uniformly continuous** if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. $|f(x) - f(y)| < \epsilon$ $\forall x, y \in I$ s.t. $|x - y| < \delta$. Note the key difference with regular continuity - here δ may only depend on ϵ only, not x.

Theorem: The Heine-Cantor Theorem

If $a < b$ and $f : [a,b] \to \mathbb{R}$ is continuous, then f is uniformly continuous.

Uniform Convergence

Definition: Pointwise Convergence

Let $E \subseteq \mathbb{R}$ be nonempty. A sequence of functions $f_n : E \to \mathbb{R}$ **converges pointwise on** E **to** $f : E \to \mathbb{R}$ if for all $x \in E$ and $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$.

Theorem: Pointwise Convergence Theorems

* The pointwise limit of bounded functions is not always bounded
* The ptwise limit of cont functs is not always cont. $\exists f_n, f : [0,1] \to \mathbb{R}$ s.t.
  - f_n is continuous for all $n \in \mathbb{N}$
  - $f_n \to f$ pointwise on [0,1] as $n \to \infty$
  - f is not continuous
  - Example: $f_n(x) := 0$ if $0 \leq x \leq \frac{1}{2} - \frac{1}{2n}$; $nx - \frac{n-1}{2}$ if $\frac{1}{2} - \frac{1}{2n} \leq x \leq \frac{1}{2} + \frac{1}{2n}$; 1 if $\frac{1}{2} + \frac{1}{2n} \leq x \leq 1$
* The pointwise limit of differentiable functions is not always differentiable. $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.:
  - f_n is differentiable for all $N \in \mathbb{N}$
  - $f_n \to f$ pointwise on $(-1,1)$ as $n \to \infty$
  - f is not differentiable
  - Example: Let $f_n(x) := \sqrt{x^2 + \frac{1}{n}}$, limit is not diffable at 0.
* The pointwise limit of derivatives is not always the derivative of the pointwise limit. $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.
  - f_n is diffable for all $n \in \mathbb{N}$ and f is diffable
  - $f_n \to f$ pointwise on $(-1,1)$
  - $f'(0) \neq \lim_{n \to \infty} f_n'(0)$, ie $\left(\lim_{n \to \infty} f_n\right)'(0) \neq \lim_{n \to \infty} f_n'(0)$ - the derivate and ptwise lim operations do not commute.
  - Example: Let $f_n(x) = \frac{\sin(2\pi n x)}{2\pi n}, f_n' = \cos(2\pi n x), f = 0$

Definition: Graph and Vertical Neighbourhood

Let $f : E \to \mathbb{R}$. The **graph of** f is the set $\gamma(f) := \{(f, f(x) : x \in E)\} \subset \mathbb{R}^2$. For $\epsilon > 0$, the $\epsilon-$vertical neighbourhood of the graph of f is the set $\mathcal{N}(f, \epsilon) := \{(x,y) : x \in E, |y - f(x)| < \epsilon\} \subset \mathbb{R}^2$.

Major Definition: Uniform Convergence

Let $E \subseteq \mathbb{R}$ be nonempty. A sequence of functions $f_n : E \to \mathbb{R}$ **converges uniformly on** E **to** $f :: E \to \mathbb{R}$ if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $x \in E, n \geq N$.
- Uniform convergence implies pointwise convergence.
  For counterexample of converse use $f_n, f : (0,1] \to \mathbb{R}$. $f_n(x) := 1$ if $0 < x < \frac{1}{n}$; 0 if $\frac{1}{n} \leq x \leq 1$ and $f(x) := 0$.

Theorem: Uniform Convergence Properties

Let $E \subseteq \mathbb{R}$ be nonempty, $f_n : E \to \mathbb{R}$ be a sequence of funcs, and $f : E \to \mathbb{R}$.
- The following are equivalent:
  (1) $f_n \to f$ uniformly on E as $n \to \infty$
  (2) $\sup_{x \in E} |f_n(x) - f(x)| \to 0$ as $n \to \infty$
  (3) For all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $\gamma(f_n) \subset \mathcal{N}(f, \epsilon)$ for all $n \geq N$.
- Let $f_n \to f$ uniformly on E as $n \to \infty$. If $(x_n)_{n \in \mathbb{N}}$ is a real sequence with $x_n \in E$ for all $n \in \mathbb{N}$, then $|f_n(x_n) - f(x_n)| \to 0$ as $n \to \infty$.

Definition: Uniform Cauchy

Let $E \subseteq \mathbb{R}$ be nonempty and $f_n : E \to \mathbb{R}$ be a sequence of functions. We say $(f_n)_{n \in \mathbb{N}}$ is **uniformly Cauchy on** E if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f_m(x)| < \epsilon$ for all $x \in E$ and $m, n \geq N$.

Major Theorem: The Uniform Cauchy Criterion

$(f_n)_{n \in \mathbb{N}}$ converges unifly on E iff $(f_n)_{n \in \mathbb{N}}$ is unifly Cauchy on E.

Thm: Uniform Limits Of Continuous/Diffable Functions

- A unif lim of diffable funcs is not always diffable. $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.:
  - f_n is differentiable for all $n \in \mathbb{N}$
  - $f_n \to f$ uniformly on $(-1,1)$ as $n \to \infty$
  - f is not differentiable
  - Example: consider $f_n(x) := \sqrt{x^2 + \frac{1}{n}}$
- Let $I \subseteq \mathbb{R}$ be a bnded open interval, $f_n : I \to \mathbb{R}$ a seq of diffable funcs s.t.:
  (1) There exists $x_0 \in I$ s.t. $(f_n(x_0))_{n \in \mathbb{N}}$ converges
  (2) There exists $g : I \to \mathbb{R}$ s.t. $f_n' \to g$ uniformly on I as $n \to \infty$.
  Then there exists diffable $f : I \to \mathbb{R}$ s.t. $f_n \to f$ uniformly on I as $n \to \infty$. Alternatively $(\lim_{n \to \infty} f_n)'(x) = \lim_{n \to \infty} f_n'(x)$ for all $x \in I$, ie differentiation commutes with taking the uniform limit under our conditions.

Definition: Uniform Convergence Of Series Of Functions

Let $E \subseteq \mathbb{R}$ be $\neq \emptyset$, $f_n : E \to \mathbb{R}$ be a seq. of real funcs. The **sequence of partial sums** is $s_n : E \to \mathbb{R}$, $s_n(x) := \sum_{k=1}^{n} f_k(x)$ for all $x \in E, n \in \mathbb{N}$. Then:
(1) $\sum_{n=1}^{\infty} f_n$ **converges ptwise on** E if $(s_n)_{n \in \mathbb{N}}$ converges ptwise on E.
(2) $\sum_{n=1}^{\infty} f_n$ **converges unifly on** E if $(s_n)_{n \in \mathbb{N}}$ converges unifly on E
(3) $\sum_{n=1}^{\infty} f_n$ **converges absolutely (pointwise) on** E if $\sum_{n=1}^{\infty} |f_n(x)|$ converges $\forall x \in E$.

Theorem: On Termwise Differentiation

Let $I \subset \mathbb{R}$ be a bdd open interval and $f_n : I \to \mathbb{R}$ a seq. of diffable funcs s.t.:
(1) There exists $x_0 \in I$ such that $\sum_{n=1}^{\infty} f_n(x_0)$ converges
(2) $\sum_{n=1}^{\infty} f_n'$ converges uniformly on I.
Then there exists diffable $f : I \to \mathbb{R}$ such that $\sum_{n=1}^{\infty} f_n$ converges uniformly to f on I, and $f'(x) = \left(\sum_{n=1}^{\infty} f_n\right)'(x) = \sum_{n=1}^{\infty} f_n'(x)$ for all $x \in I$.

Theorem: Weierstrass M-Test

Let $E \subseteq \mathbb{R}$ be $\neq \emptyset$, $f_n : E \to \mathbb{R}$, and $(M_n)_{n \in \mathbb{N}}$ a non-negative seq satisfying:
(1) $|f_n(x)| \leq M_n$ for all $x \in E, n \in \mathbb{N}$
(2) $\sum_{n=1}^{\infty} M_n$ converges
Then $\sum_{n=1}^{\infty} f_n$ converges absolutely and uniformly on E.
Proof. Let $\epsilon > 0$. By the CC for series, $\exists N \in \mathbb{N}$ s.t. $\sum_{k=m+1}^{n} M_k < \epsilon$ for $n \geq m \geq N$. Thus by hypothesis, $\left|\sum_{k=m+1}^{n} f_k(x)\right| \leq \sum_{k=m+1}^{n} |f_k(x)| \leq \sum_{k=m+1}^{n} M_k < \epsilon$ for all $x \in E, n > m \geq N$. Thus the partial sums of $\sum_{n=1}^{\infty} f_n$ are unifly Cauchy, so $\sum_{n=1}^{\infty} f_n$ converges unifly. Moreover, by the above, the partial sums of $\sum_{n=1}^{\infty} |f_n|$ are unifly Cauchy, so the series converges absolutely. □

Theorem: Continuity Of Uniform Limits

Let $I \subseteq \mathbb{R}$ be an interval, $f_n, f : I \to \mathbb{R}$ s.t. $f_n \to f$ unifly on I as $n \to \infty$. For all $x_0 \in I$, if each f_n is continuous at x_0, then f is continuous at x_0.
Proof. Let $\epsilon > 0$. Since $f_n \to f$ unifly on I, $\exists N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon/3$ for all $x \in I, n \geq N$. Since f_N is cont. at $x_0 \in I$, $\exists \delta > 0$ s.t. $|f_N(x) - f_N(x_0)| < \epsilon/3$ for all $x \in I$ s.t. $|x - x_0| < \delta$. Now let $x \in I$ be s.t. $|x - x_0| < \delta$. Then by the TI, $|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \epsilon$. Thus f is cont. at $x_0 \in I$. □

Power Series

Definition: Power Series

Let $(a_n)_{n \in \mathbb{N}}$ be a real sequence and $c \in \mathbb{R}$.
A **power series centred at** c is a series of the form $\sum_{n=0}^{\infty} a_n (x - c)^n$.

Definition: Radius Of Convergence

Let $(a_n)_{n\in\mathbb{N}}$ be a real seq and $c\in\mathbb{R}$. Then one of the following holds:
(1) $\sum_{n=0}^{\infty} a_n(x-c)^n$ converges only when $x=c$
(2) $\sum_{n=0}^{\infty} a_n(x-c)^n$ converges absolutely for all $x\in\mathbb{R}$
(3) There exists $R>0, R\in\mathbb{R}$ such that
$\sum_{n=0}^{\infty} a_n(x-c)^n$ converges absolutely when $|x-c|<R$ and diverges when $|x-c|>R$. Here $R=\sup\{r>0:(a_n r^n)_{n\in\mathbb{N}_0}$ is bounded $\}$

The R in case (3) is the **radius of convergence** of $\sum_{n=0}^{\infty} a_n(x-c)^n$. In cases (1) and (2) we say $R=0$ and $R=\infty$ respectively.

Theorem: On Termwise Differentiation Of Power Series

Let $\sum_{n=0}^{\infty} a_n(x-c)^n$ have radius of convergence $0<R\leq\infty$.
Then $f(x)=\sum_{n=0}^{\infty} a_n(x-c)^n$ for $x\in(c-R,c+R)$ is differentiable on $(c-R,c+R)$; and $f'(x)=\sum_{n=1}^{\infty} na_n(x-c)^{n-1}$ for all $x\in(c-R,c+R)$.

- Given the above, $f(x):=\sum_{n=0}^{\infty} a_n(x-c)^n$ is **infinitely differentiable** on $(c-R,c+R)$, and $a_k=\frac{f^{(k)}(c)}{k!}$ for all $k\in\mathbb{N}_0$.
- For $0<r<R$, both series converge absolutely and unifly on $[c-r,c+r]$
- For any real sequence $(a_n)_{n\in\mathbb{N}_0}$ and $c\in\mathbb{R}$, $\sum_{n=0}^{\infty} a_n(x-c)^n$ and $\sum_{n=1}^{\infty} na_n(x-c)^{n-1}$ have the same radius of convergence

Major Theorem: Uniqueness Of Power Series

Power series are unique - For $\sum_{n=0}^{\infty} a_n(x-c)^n$ and $\sum_{n=0}^{\infty} b_n(x-c)^n$ with RoCs $0<R,S,\leq\infty$, if $\exists 0<r<\min R,S$ s.t. $\sum_{n=0}^{\infty} a_n(x-c)^n=\sum_{n=0}^{\infty} b_n(x-c)^n$ for all $x\in\mathbb{R}$ s.t. $|x-c|<r$, then $a_n=b_n$ for all $n\in\mathbb{N}$.

Proof. Let $I=(c-r,c+r)$, and $f:I\to\mathbb{R}, f(x):=\sum_{n=0}^{\infty} a_n(x-c)^n$ for $x\in I$. We know f is infinitely diffable on I, and $a_n=\frac{f^{(n)}(c)}{n!}$ for $n\in\mathbb{N}_0$. However, $f(x)=\sum_{n=0}^{\infty} b_n(x-c)^n$ for $x\in I$, and $b_n=\frac{f^{(n)}(c)}{n!}$ for all $n\in\mathbb{N}_0$. Thus $a_n=b_n$ for all $n\in\mathbb{N}_0$. □

Definition: The Exponential

Consider $\sum_{n=0}^{\infty}\frac{x^n}{n!}$ with RoC $R=\infty$. $\exp(x):=e^x:=\sum_{n=0}^{\infty}\frac{x^n}{n!}$.

Definition: Taylor Series

Let $I\subseteq\mathbb{R}$ be a nonempty open interval, and $f:I\to\mathbb{R}$ be infinitely diffable. Then the **Taylor series of** f **at** $c\in I$ is $\sum_{n=0}^{\infty}\frac{f^{(n)}(c)}{n!}(x-c)^n$.
f is **real analytic** if its TS converges ptwise on I and equals $f(x)$ at x

Measure And Integration

The Lebesgue Measure

Definition: The Lebesgue Outer Measure

For $E\subseteq\mathbb{R}$, the **Lebesgue outer measure of** E is $m^*(E):\mathcal{P}(\mathbb{R})\to[0,\infty]:=\inf\left\{\sum_{j=1}^{\infty}\lambda(I_j):E\subseteq\bigcup_{j=1}^{\infty}I_j, I_j\subseteq\mathbb{R}\text{ an open interval}\right\}$. Here the inf is over all countable coverings $(I_j)_{j\in\mathbb{N}}$ of E by open intervals.

* $m^*(\emptyset)=0$
* If $E\subseteq F\subseteq\mathbb{R}$, $m^*(E)\leq m^*(F)$
* If $(E_j)_{j\in\mathbb{N}}$ is any seq of subsets of $\mathbb{R}$, $m^*\left(\bigcup_{j=1}^{\infty}E_j\right)\leq\sum_{j=1}^{\infty}m^*(E_j)$
- If $\sum_{j=1}^{\infty}\lambda(I_j)$ diverges, we say it has value ∞. If there is no cover of E s.t. $\sum_{j=1}^{\infty}\lambda(I_j)<\infty$, $m^*(E)=\infty$.
- For any interval $J\subseteq\mathbb{R}$, $\lambda(J)=m^*(J)$.
- Given $E\subseteq\mathbb{R}$, $\epsilon>0$, $\exists$ open $U\subseteq\mathbb{R}$ s.t. $E\subseteq U$ and $m^*(U)<m^*(E)+\epsilon$.

Definition: Null Sets

A set $E\subseteq\mathbb{R}$ is **null** or **has measure zero** if $m^*(E)=0$.
Every countable subset of R has measure zero.

Proof. First prove any singleton set $\{x\}$ for $x\in\mathbb{R}$ has measure zero. Given $x\in\mathbb{R}$, let $\epsilon>0$ and consider the cover $I:=(x-\epsilon/2,x+\epsilon/2)$. Then $m^*(\{x\})\leq\lambda(I)=\epsilon$. Since $\epsilon>0$ is arbitrary we conclude $m^*(\{x\})=0$. Any countable E is a countable union of singletons, $E=\bigcup_{j=1}^{\infty}\{x_j\}$.
Apply the countable subadditivity property to conclude $m^*(E)=0$. □

Definition: Distance Between Sets

For $E,F\subseteq\mathbb{R}\neq\emptyset$ and disjoint; $\operatorname{dist}(E,F):=\inf\{|x-y|:x\in E,y\in F\}$.
- If $E,F\subseteq\mathbb{R}\neq\emptyset$, disjoint, $\operatorname{dist}(E,F)>0$; $m*(E\cup F)=m^*(E)+m^*(F)$
- If $F<K\subseteq\mathbb{R}$ nonempty, closed, $F\cup K=\emptyset$, K bounded; $\operatorname{dist}(F,K)>0$.
- For $(K_j)_{j\in\mathbb{N}}$ a sequence of disjoint closed $K_i\subseteq\mathbb{R}$; $m^*\left(\bigcup_{j=1}^{\infty}K_j\right)=\sum_{j=1}^{\infty}m^*(K_j)$

Major Definition: Lebesgue Measurable Sets

$E\subseteq\mathbb{R}$ is **Lebesgue measurable**
if for all $\epsilon>0$, there exists open $U\subseteq\mathbb{R}$ s.t. $E\subseteq U$ and $m^*(U\setminus E)<\epsilon$.
The collection of all L-mesable subsets of $\mathbb{R}$ is denoted $\mathcal{M}$.

Theorem: Properties Of L-Measurable Sets

- Let $F\subseteq\mathbb{R}$ and $(E_j)_{j\in\mathbb{N}}$ be L-mesable. Then:
  (1) The union $E:=\bigcup_{j=1}^{\infty}E_j$ is L-mesable
  (2) The complement $\mathbb{R}\setminus F$ is L-mesable
  (3) The intersection $\bigcap_{j=1}^{\infty}E_j$ is L-mesable
- Let $E,F\subseteq\mathbb{R}$ be L-mesable. Then $F\setminus E$ is L-mesable
- For $E\subseteq\mathbb{R}$, E is L-mesable iff $\forall\epsilon>0$, $\exists$ closed $K\subseteq E$ and $m^*(E\setminus K)<\epsilon$.

Major Definition: The Lebesgue Measure

The **Lebesgue measure**
is the function $m:\mathcal{M}\to[0,\infty]$ given by $m(E):=m^*(E)$ for $E\in\mathcal{M}$.

Theorem: Countable Additivity Of The Lebesgue Measure

For disjoint L-mesable $(E_j)_{j\in\mathbb{N}}$, $m\left(\bigcup_{j=1}^{\infty}E_j\right)=\sum_{j=1}^{\infty}m(E_j)$.

Theorem: Monotone Convergence For Measure

Let $(E_j)_{j\in\mathbb{N}}$ be a seq. of L-mesable sets s.t. $E_j\subseteq E_{j+1}$ for all $j\in\mathbb{N}$. Then $m\left(\bigcup_{j=1}^{\infty}E_j\right)=\lim_{j\to\infty}E_j$

The Lebesgue Integral

Definition: Characteristic Functions Of Sets

For $E\subseteq\mathbb{R}$, the **characteristic function of** E is
$\chi_E:\mathbb{R}\to\mathbb{R}$, $\chi_E(x):=1$ if $x\in E$; $:=0$ if $x\notin E$.

Definition: Simple Functions

$s:\mathbb{R}\to\mathbb{R}$ is **simple** if it is a linear combination of characteristic functions of L-mesable sets. Ie, $\exists c_1,\ldots,c_k\in\mathbb{R}$ and L-mesable $E_1,\ldots,E_k\subseteq\mathbb{R}$ s.t. $s(x)=\sum_{j=1}^{k}c_j\chi_{E_j}(x)$ for all $x\in\mathbb{R}$.

Definition: L-Integral Of A Non-Negative Simple Function

Let $s:\mathbb{R}\to[0,\infty)$ be simple, $a_j\in[0,\infty), E_j\in\mathcal{M}$, $s=\sum_{j=1}^{k}a_j\chi_{E_j}$.
The **Lebesgue integral of** s is $\int_{\mathbb{R}}s(x)\ \mathrm{d}x:=\sum_{j=1}^{k}a_j m(E_j)\in[0,\infty)$.
- $\int_{\mathbb{R}}\chi_E(x)\ \mathrm{d}x:=m(E)$
- This L-integral respects linearity

Major Definition: Lebesgue Measurable Functions

Let $E\subseteq\mathbb{R}$ be L-mesable. $f:\mathbb{E}\to\mathbb{R}$ is **Lebesgue measurable**
if for all $\lambda\in\mathbb{E}$, the strict superlevel set $\{x\in\mathbb{E}:f(x)>\lambda\}$ is L-mesable
- Any simple function is L-mesable
- Any continuous function $f:\mathbb{R}\to\mathbb{R}$ is L-mesable
- If $E\subseteq\mathbb{R}$, $f,g:\mathbb{R}\to\mathbb{R}$ are all L-mesable; then $\max\{f,g\}$, $\min\{f,g\}$, $f\cdot\chi_E$ are all L-mesable
- If $f,g:\mathbb{R}\to\mathbb{R}$ L-mesable, $\alpha,\beta\in\mathbb{R}$; then $\alpha f+\beta g$, $f\cdot g$ are L-mesable.

Theorem: Approxing Non-Negative L-Measurable Functions

Let $f:\mathbb{R}\to[0,\infty)$ be a non-negative L-mesable func. Then there exists a sequence $s_n:\mathbb{R}\to[0,\infty)$ of non-negative simple functions s.t.:
(1) $(s_n)_{n\in\mathbb{N}}$ is ptwise non-decreasing - $0\leq s_n\leq s_{n+1}$ for $x\in\mathbb{R}$, $n\in\mathbb{N}$.
(2) $s_n\to f$ pointwise on $\mathbb{R}$ as $n\to\infty$.

Dfn: Integrating Non-Negative L-Measurable Functions

Let $f:\mathbb{R}\to[0,\infty)$ be a non-negative L-mesable func. The **Lebesgue integral of** f as $\int_{\mathbb{R}}f(x)\ \mathrm{d}x:=\sup\{\int_{\mathbb{R}}s(x)\ \mathrm{d}x:s\text{ simple }, 0\leq s\leq f\}$ where the supremum is taken over all non-negative simple functions $s:\mathbb{R}\to\mathbb{R}$ with $s(x)\leq f(x)$ for all $x\in\mathbb{R}$.
- If $f:\mathbb{R}\to[0,\infty)$, $E\subseteq\mathbb{R}$ both L-mesable, $\int_E f(x)\ \mathrm{d}x:=\int_{\mathbb{R}}f(x)\chi_E(x)\ \mathrm{d}x$
- For L-mesable f defined only on L-mesable $E\subseteq\mathbb{R}$, define $\int_E f(x)\ \mathrm{d}x=\int_{\mathbb{R}}\tilde{f}(x)\ \mathrm{d}x$, with $\tilde{f}:\mathbb{R}\to[0,\infty)$, $\tilde{f}(x):=f(x)$ if $x\in E$, 0 if $x\in\mathbb{R}\setminus E$.
- Let $f,g:\mathbb{R}\to[0,\infty)$ be L-mesable. Then:
  (1) If $0\leq f(x)\leq g(x)$ for all $x\in\mathbb{R}$, $0\leq\int_{\mathbb{R}}f(x)\ \mathrm{d}x\leq\int_{\mathbb{R}}g(x)\ \mathrm{d}x$
  (2) For all $\lambda>0$, $m(\{x\in\mathbb{R}:f(x)>\lambda\})\leq\frac{1}{\lambda}\int_{\mathbb{R}}f(x)\ \mathrm{d}x$
- This L-integral respects linearity
- If $f:\mathbb{R}\to[0,\infty)$ is L-mesable and $E,F\subseteq\mathbb{R}$ are disjoint L-mesable sets, $\int_{E\cup F}f(x)\ \mathrm{d}x=\int_E f(x)\ \mathrm{d}x+\int_F f(x)\ \mathrm{d}x$
- Let $h_k:\mathbb{R}\to[0,\infty)$ be a seq of non-neg L-mesable funcs on $\mathbb{R}$ s.t. $\sum_{k=1}^{\infty}h_k$ cnvges ptwise. Then $\sum_{k=1}^{\infty}\int_{\mathbb{R}}h_k(x)\ \mathrm{d}x=\int_{\mathbb{R}}\sum_{k=1}^{\infty}h_k(x)\ \mathrm{d}x$.

Theorem: Pointwise Limits And L-Integration

The pointwise limit of L-integrals is not necessarily the L-integral of the pointwise limit. Ie, $\exists$ L-mesable $f_n,f:\mathbb{R}\to[0,\infty)$ s.t.:
- f_n is L-integrable for all $n\in\mathbb{N}$ and f is L-integrable
- $f_n\to f$ pointwise on $(0,1]$
- $\int_0^1 f(x)\ \mathrm{d}x\neq\lim_{n\to\infty}\int_0^1 f_n(x)\ \mathrm{d}x$, ie $\lim_{n\to\infty}\int_0^1 f_n(x)\ \mathrm{d}x\neq\int_0^1(\lim_{n\to\infty}f_n(x))\ \mathrm{d}x$
- For counterexample: $f,f_n:\mathbb{R}\to[0,\infty)$, $f(x):=\mathbb{R}\to[0,\infty)$, $f_n(x):=n$ if $0<x\leq\frac{1}{n}$ and 0 otherwise.

Major Theorem: The MCT For Lebesgue Integrals

Let $f_k:\mathbb{R}\to[0,\infty)$ be a ptwise non-decreasing seq. of non-negative L-mesable funcs converging ptwise to $f:=\lim_{k\to\infty}f_k:\mathbb{R}\to[0,\infty)$. Then f is L-mesable $\lim_{k\to\infty}\int_{\mathbb{R}}f_k(x)\ \mathrm{d}x=\int_{\mathbb{R}}\lim_{k\to\infty}f_k(x)\ \mathrm{d}x=\int_{\mathbb{R}}f(x)\ \mathrm{d}x$.

Lemma: The Fatou Lemma

If $f_k:\mathbb{R}\to[0,\infty)$ is a seq of non-neg L-mesable funcs, then $\int_{\mathbb{R}}\liminf f_{k\to\infty}f_k(x)\ \mathrm{d}x\leq\liminf_{k\to\infty}\int_{\mathbb{R}}f_k(x)\ \mathrm{d}x$.

Lemma: The Reverse Fatou Lemma

Let $f_k:\mathbb{R}\to[0,\infty)$ be a seq of non-neg L-mesable funcs and $g:\mathbb{R}\to[0,\infty)$ L-intable s.t. $f_k(x)\leq g(x)$ for all $x\in\mathbb{R}$. Then $\int_{\mathbb{R}}\limsup_{k\to\infty}f_k(x)\ \mathrm{d}x\geq\limsup_{k\to\infty}\int_{\mathbb{R}}f_k(x)\ \mathrm{d}x$.

Dfn: Decomposing L-Measurable Real-Valued Functions

Let $f:\mathbb{R}\to\mathbb{R}$ be L-mesable. $f^+(x):=\max\{f(x),0\}$ and $f^-(x):=-\min\{f(x),0\}$, $f=f^+-f^-$, and $|f|=f^++f^-$ are all L-mesable. Moreover, $\int_{\mathbb{R}}|f(x)|\ \mathrm{d}x<\infty\iff\int_{\mathbb{R}}f^+(x)\ \mathrm{d}x<\infty$ and $\int_{\mathbb{R}}f^-(x)\ \mathrm{d}x<\infty$.

MDfn: Lebesgue Integration Of Real-Valued Functions

An L-mesable $f:\mathbb{R}\to\mathbb{R}$ is **Lebesgue integrable** if $\int_{\mathbb{R}}|f(x)|\ \mathrm{d}x<\infty$.
The collection of all L-integrable functions is $\mathcal{L}^1(\mathbb{R})$.
For L-integrable $f:\mathbb{R}\to\mathbb{R}$, $\int_{\mathbb{R}}f(x)\ \mathrm{d}x:=\int_{\mathbb{R}}f^+(x)\ \mathrm{d}x-\int_{\mathbb{R}}f^-(x)\ \mathrm{d}x$.
- For $f\in\mathcal{L}^1(\mathbb{R})$, $\left|\int_{\mathbb{R}}f(x)\ \mathrm{d}x\right|\leq\int_{\mathbb{R}}|f(x)|dx$
- This L-integral respects linearity
- For L-mesable $E\subseteq\mathbb{R}$, $\mathcal{L}^1(E)=\left\{f:E\to\mathbb{R}:f\text{ is L-mesable, }\tilde{f}\in\mathcal{L}^1(\mathbb{R})\right\}$.

Major Theorem: The Dominated Convergence Theorem

Let $f_k:\mathbb{R}\to\mathbb{R}$ be a seq. of L-integrable funcs converging ptwise to $f:\mathbb{R}\to\mathbb{R}$, and $g\in\mathcal{L}^1(\mathbb{R})$ be s.t. $|f_k(x)|\leq g(x)$ for all $x\in\mathbb{R}, k\in\mathbb{N}$. Then $f\in\mathcal{L}^1(\mathbb{R})$ and $\lim_{k\to\infty}\int_{\mathbb{R}}f_k(x)\ \mathrm{d}x=\int_{\mathbb{R}}\lim_{k\to\infty}f_k(x)\ \mathrm{d}x=\int_{\mathbb{R}}f(x)\ \mathrm{d}x$.
- Uniform Convergence Theorem for integrals: Let $E\subseteq\mathbb{R}$ be L-mesable s.t. $m(E)<\infty$, and $f_n:\mathbb{R}\to\mathbb{R}$ a seq. of L-integrable funcs s.t. $f_n\to f$ unifle on E as $n\to\infty$, for $f:\mathbb{R}\to\mathbb{R}$. Then $f\in\mathcal{L}^1(\mathbb{R})$ and $\lim_{n\to\infty}\int_E f_n(x)\ \mathrm{d}x=\int_E\lim_{n\to\infty}f_n(x)\ \mathrm{d}x$

Theorem: The Fundamental Theorem Of Calculus

Let $a,b\in\mathbb{R}$ with $a<b$ and $f:[a,b]\to\mathbb{R}$ continuous. Then $F(x):=\int_a^x f(y)\ \mathrm{d}y$ is diffable on $[a,b]$ and $F'(x)=f(x)$ for all $x\in[a,b]$
- For $a,b\in\mathbb{R}$, $a<b$, $f:[a,b]\to\mathbb{R}$ cont.sly diffable; $\int_a^b f'(x)\ \mathrm{d}x=f(b)-f(a)$